



Watermarking Spatial method

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Outline

Introduction

Embedding

Decoding

Outline

Introduction

Embedding

Decoding

Introduction

Your ?

?



?



My ?

?

?

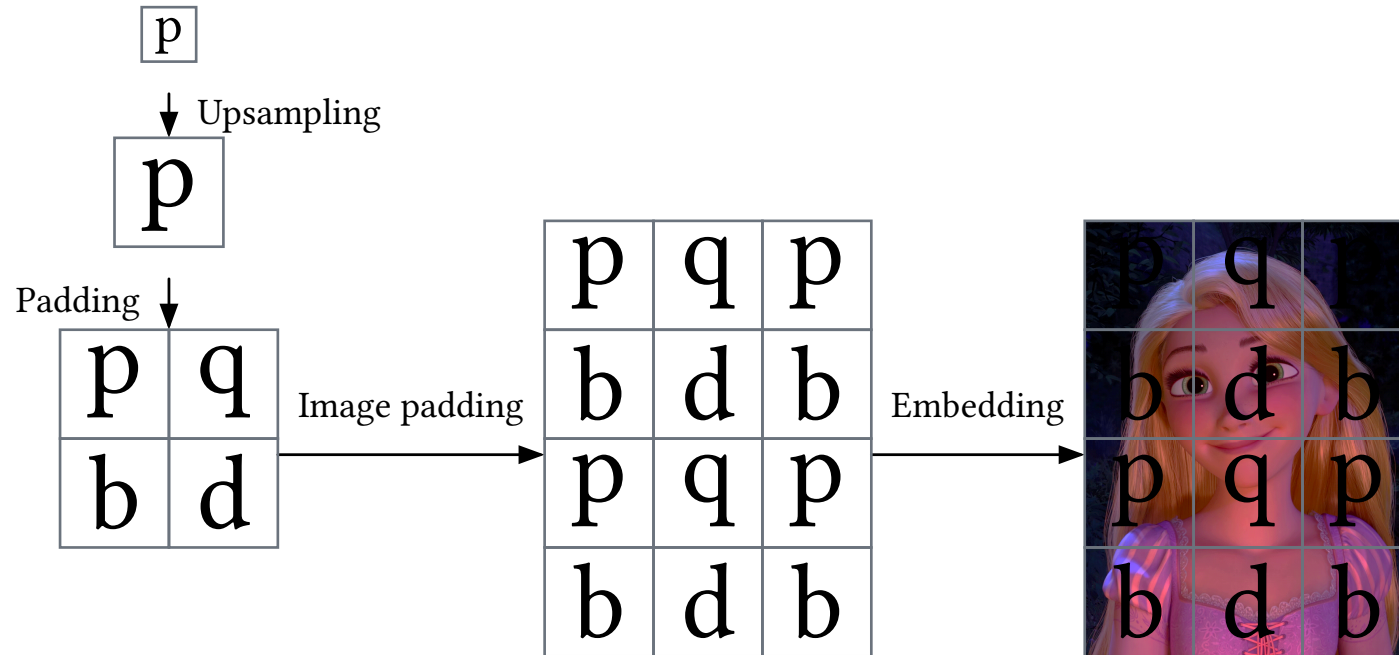
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Embedding



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Decoding

Decoding process

- Denoising
- Direct decoding (baseline)
- Synchronization
- Patch extraction
- Message recovery
- Return best of both decoding paths

Decoding

Synchronization process

- Autocorrelation
- Peak detection
- Select reference peaks
- Get rotation and scaling parameters
- Apply inverse transform
- Get translation and patch orientation
- Apply inverse transform

Decoding

Patch extraction process

- Divide residual into patches
- Sum up patches
- positive = 1, negative = 0
- Get confidence score (mean of absolute values in patch)
- Return bit sequence and confidence scores